From YouTube to Reddit: Topic and Emotion Dynamics in World of Warcraft*

Jyoti Kanwar Rajawat, Elena Kim and Jaewon Choi
Soonchunhyang University, South Korea
{Rajawat, voogiejam22, jaewonchoi}@sch.ac.kr

Abstract

We investigate cross-platform topic and sentiment dynamics between YouTube and Reddit using World of Warcraft as a case. Based on 119,871 comments (2020–2025), we apply LDA, dynamic topic modeling (DTM), VADER sentiment analysis, social network analysis (SNA), and a multilayer perceptron (MLP). Results show that YouTube sentiment and topic intensity are significantly associated with subsequent Reddit discussion activity, explaining 67% of the variance in comment volume. We further identify a consistent diffusion pattern in which emotionally expressive content on YouTube precedes more structured discussions on Reddit, with an average delay of approximately 55–60 hours. The study contributes by linking media richness theory with cross-platform behavioral modeling and provides practical implications for data-driven community management and digital marketing.

1 Introduction

According to a new report by Newzo (2024), the global online gaming market is projected to grow from $182.9 billion in 2022 to more than $213.3 billion by 2027, reflecting the rapid growth of the gaming industry and its more than 3.3 billion global user bases. As the online gaming industry continues to grow, the way players interact and communicate continues to evolve. The rise of social media and online communities has enabled players to go beyond in-game communication and actively share their gaming experiences, discuss content updates, express their emotional reactions, and engage in the collective construction of meaning in a broader online space.

Drawn from the Media Richness Theory [7] , social media platforms such as YouTube are high richness media, capable of providing multiple information cues and supporting complex emotional

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expressions; in contrast, communities such as Reddit, although also interactive, are more often used as vehicles for structured knowledge exchange and are closer to the characteristics of low richness media. Meanwhile, Media Synchronization Theory [10] points out that there are also differences in synchronization across platforms: high synchronization platforms are more suitable for information storage and asynchronous communication.

Despite the growing importance of platform-based communication in contemporary gaming culture, most existing research on gaming communities focuses on single-platform or static topic analyses, with less coverage of the dynamic evolution of cross-platform topics, and, in particular, there is still a gap in research on predicting the future activity of professional gaming communities. This gap is particularly noteworthy because the modern social media ecosystem is highly intertwined, with players switching frequently between platforms: they may first watch tutorial videos on YouTube, browse the comment section, express their opinions, and then turn to professional communities like Reddit for help or in-depth discussion.

Taking World of Warcraft (WoW) as a case study, this research aims to investigate how topics and emotions evolve between YouTube and Reddit over a 5 year period of time. This study addresses the following questions:

RQ1: How do player-discussed topics evolve dynamically across YouTube and Reddit?

RQ2: What are the dominant diffusion patterns and time delays between these platforms?

RQ3: Can YouTube sentiment and topic intensity effectively predict future activity levels in the Reddit community?

This study makes two specific contributions. First, it demonstrates that sentiment and topic intensity signals from a high-reach, non-specialized platform (YouTube) explain a substantial portion of variation in discussion activity within a specialized community (Reddit), accounting for approximately 67% of the variance in comment volume. Second, it identifies a consistent cross-platform diffusion pattern in which emotionally expressive content precedes structured, knowledge-oriented discussions, thereby linking media richness theory with observable behavioral dynamics in online gaming ecosystems

2 Literature review

2.1 Specialized vs Non-specialized Communities

Online gaming discussions occur across heterogeneous platforms that differ in community structure and user motivation. Prior research distinguishes between specialized communities and non-specialized communities, which are more spontaneous, open, and emotionally expressive [23]. Reddit exemplifies a specialized community: it organizes information through subforums, tagging, and chronological threads that facilitate structured, knowledge-intensive exchanges. Users often display functional motivations, such as problem-solving, game optimization, and strategic analysis [1]. By contrast, YouTube functions as a non-specialized community emphasizing affective engagement. Its comment sections and video responses allow users to express immediate emotional reactions and social identification without requiring deep expertise. In the World of Warcraft ecosystem, these contrasting features shape both the depth and pace of information diffusion. Reddit users are primarily driven by functional motivations such as problem-solving and game optimization, whereas YouTube users are more driven by social and emotional motivations, expressing reactions and identity through comments and interactions. As a result, Reddit facilitates structured knowledge accumulation and in-depth discussion, while YouTube enables rapid, large-scale emotional expression but provides limited support for systematic knowledge exchange.

2.2 Media Richness Theory and Media Synchronicity Theory

Media Richness Theory (MRT) [7] posits that communication media vary in their capacity to convey complex, nuanced information through feedback immediacy, multiple cues, language variety, and personalization [7]. High-richness media are better suited for ambiguous or emotion-laden exchanges, while low-richness media excel in precise, structured information sharing. Applied to gaming contexts, YouTube's audiovisual format and emotive tone offer high richness but limited interactivity, whereas Reddit's text-based discussions enable detailed argumentation and information storage, aligning with low-richness but cognitively richer communication. YouTube exhibits strong media richness through multimodal content, consistent with [21], its affordances are insufficient for constructing or responding to complex, knowledge-based discussions. From another perspective, [25] argue that media richness should be understood as an intrinsic and relatively stable characteristic determined by the medium's objective properties.

While MRT effectively distinguishes between communication capacities across media, it lacks a mechanism to explain how media adapt to communication processes. Media Synchronicity Theory further distinguishes between two core communication processes: conveyance (the transmission of information) and convergence (the development of shared understanding). In this context, YouTube primarily supports conveyance through rapid dissemination of audiovisual content, while Reddit facilitates convergence through threaded discussions, allowing users to refine, debate, and integrate knowledge. These differences help explain why emotionally driven content often emerges on YouTube before being elaborated into structured discussions on Reddit.

3 Hypotheses development

As gaming communities become more interconnected, users increasingly interact and produce content across multiple platforms, making platform type and communication mechanism critical variables in understanding behavioral dynamics. Within this context, Media Dependency Theory posits that an individual's reliance on a particular medium is determined by the extent to which it fulfils their needs for information, reality interpretation, and behavioral guidance [2]. In the context of gaming, these dependencies manifest differently across platforms: Reddit, as a text-centric and specialized community, facilitates rational analysis and technical discourse surrounding game updates and system adjustments, while YouTube, as a visually driven non-specialized platform, emphasizes entertainment and emotional expressions. YouTube users often communicate affective reactions and subjective viewpoints through videos, while Reddit users focus on analytical and knowledge-based exchanges. This functional differentiation profoundly shapes how WoW topics are framed and discussed across communities.

Furthermore, [11] issue attention cycle theory suggests that public attention toward a specific topic follows the sequence: "initial trigger – rapid escalation – short-lived climax – gradual decline". This cycle is especially evident in gaming discourse. Whenever WoW launches a major update, there is an immediate surge of attention and collective feedback, followed by a gradual decrease in engagement and dispersion of discussions. The intensity and duration of this cycle vary across platforms, supporting this, [24] found notable cross-platform differences in player discussions of "Among Us", highlighting the role of platform attributes in topic formation and evolution. Similarly, Mitrovic et al. (2025) demonstrated through comparative analysis of Reddit that users' content creation and interaction behaviors display clear structural variations across platforms, offering methodological support for cross-platform analysis. Hence, this study collects user comment data on the WoW "game update" topic

to examine the temporal evolution and platform-based dynamics of discussions. Accordingly, we propose the following hypothesis:

H1: Game topics discussed by players evolve dynamically over time, and the pattern of change differs by platform type.

H1 is tested using Dynamic Topic Modeling (DTM) by dividing the dataset into three time phases (Emergence, Growth, Saturation). Topic contribution scores are compared across phases and platforms. Evidence supporting H1 is indicated by (1) changes in dominant topics over time and (2) differences in topic prominence between YouTube and Reddit.

Game-related topics often exhibit clear cross-platform communication patterns, where information dissemination occurs collaboratively through interlinked individual, group, and platform layers [17]. User behavior is not only shaped by platform-specific rules and community cultures but also contributes to broader cross-platform interaction paths within this networked environment. [20] identified that active users across multiple platforms often serve as bridge nodes, facilitating the flow of information between spaces such as Reddit and Twitter. In gaming contexts, players commonly watch and comment on videos via YouTube, then migrate discussions to Reddit for more in-depth technical elaboration or critical feedback. The mechanism called migratory dissemination captures how content circulates from non-specialized to specialized communities. Thus, the second hypothesis is formulated:

H2: Popular game topics tend to spread from non-specialized social media (YouTube) to specialized gaming communities (Reddit), and the spreading paths show specific network structure characteristics.

H2 is tested by constructing a topic-level communication network linking semantically similar topics across platforms using cosine similarity and temporal proximity. Cross-platform diffusion is operationalized as the time difference between the first appearance of a topic on YouTube and its subsequent appearance on Reddit. Support for H2 is indicated by consistent directional delays and centrality patterns in the topic network.

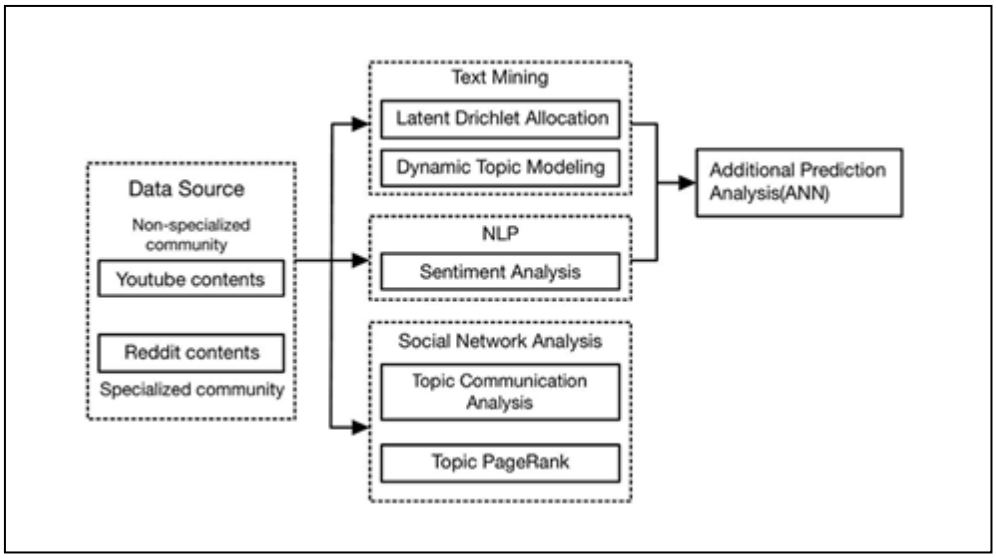
Within multi-platform ecosystems, users' motivations and emotional expressions become essential indicators for understanding topic diffusion and platform-level role differentiation. User motivation comprises cognitive, emotional, and behavioral dimensions [15] on platforms like YouTube, the forms of expression foster both cognitive attitudes and shared emotional climates through affective commenting, which in turn precedes the rational analysis typically observed on Reddit. Recent studies reinforce the predictive potential of sentiment in cross-platform contexts. [28] highlight that sentiment analysis across social media can effectively forecast discussion activity and interaction trajectories. In gaming environments characterized by high emotional density, user sentiment is concentrated and volatile, serving as a leading indicator of opinion dynamics. Lin et al. (2024) distinguish between emotional tendency, which predicts the breadth of discussion, and emotional volatility, which signals potential surges in topic activity. On a non-specialized platform such as YouTube, the high volume and emotional expressiveness of comments provide early cues for predicting subsequent topic growth on specialized forums. Thus, we propose the following hypothesis:

H3: Sentiment trends and topic heats on a non-specialized platform (YouTube) can significantly predict future discussion activity in a specialized community (Reddit).

H3 is tested using a supervised learning model in which aggregated YouTube features (VADER sentiment scores, TF-IDF–based topic intensity, and interaction metrics such as likes and replies) are used to explain Reddit comment volume at the same temporal aggregation level (daily). Model performance is evaluated using Mean Squared Error (MSE) and R^2. As the model uses contemporaneous features, the analysis focuses on explanatory relationships rather than strict temporal forecasting.

4 Research Methodology

4.1 Research Framework



Error! Reference source not found.. Research framework

This study investigates the phenomenon of cross-platform thematic diffusion in the World of Warcraft community, focusing on interactions between YouTube and Reddit. The analysis integrates text mining, natural language processing (NLP), and social network analysis (SNA) to examine how sentiments and thematic patterns emerging on YouTube influence subsequent. The research framework (figure 1) is designed to capture the multi-stage analytical process.

4.2 Data Collection and Preprocessing

Data were collected from YouTube and Reddit using their official APIs. For YouTube, comments were retrieved via the YouTube Data API by querying keywords related to World of Warcraft, including "World of Warcraft", "WoW update", and expansion-specific terms such as "Shadowlands" and "Dragonflight". We selected videos with high engagement (e.g., high view counts and active comment sections) to ensure sufficient user interaction data.

For Reddit, data were collected using the Pushshift API from relevant subreddits, including r/wow and r/classicwow. These communities were selected because they represent major hubs for World of Warcraft discussions and technical exchange.

The dataset spans from January 2020 to May 2025. After initial retrieval, duplicate entries and non-relevant comments were removed. The final dataset consists of 119,871 comments, including 79,844 from YouTube and 40,027 from Reddit. To align the analysis with major game lifecycle events, data collection covers multiple expansion and patch periods of World of Warcraft. These periods provide natural variation in user attention and discussion intensity, which supports the analysis of topic evolution and cross-platform dynamics.

Text preprocessing included removing URLs, emojis, and non-textual elements, followed by tokenization and stopword removal. Only English-language comments were retained for analysis.

5 Results

5.1 Topic modeling (LDA)

We adopt Latent Dirichlet Allocation (LDA) a Bayesian generative model for unsupervised topic discovery in documents [5]. In LDA, each document di is a mixture of K topics and each topic is a distribution over the vocabulary ,LDA topic analysis was also performed on the comment data in the Reddit platform, and num-topics were also set to 5 to identify and extract potential topic patterns in the text. For each theme, the top 8 words with the highest probability were extracted as the core representation words of the theme, and the theme modeling results of World of Warcraft comments on Reddit. To summarize, Reddit users show higher interactivity and technicality in World of Warcraft topics, focusing more on in-depth exchanges of game experience and diversified discussions of game mechanics. These results indicate clear differences in topic focus across platforms. Reddit discussions are more technical and interaction-driven, emphasizing gameplay strategies and problem-solving, while YouTube discussions are more descriptive and experience-oriented.

5.2 Dynamic Topic modeling (DTM)

This study further explores the potential changes in textual content over the time series by performing Dynamic Theme Analysis (DTM) on the comment data related to World of Warcraft in the YouTube platform. Specifically, the entire dataset is first divided into three phases in chronological order, i.e., Emergence, Growth, and Saturation, in order to observe the dynamic evolution of user discussions in the time dimension. In each period, the number of topics (num-topics) in the LDA model is set to 5 to ensure the consistency and comparability of topic division. For each topic, the top 8 high-frequency words are extracted to show the hotspots of users' attention at different stages. We also calculate and average the contribution scores of the topic terms for each period in YouTube platform to better compare the importance and influence of the topics at different stages. The average contribution score distribution of topics in different time periods can be visualized through the line graph in Figure2 and figure 3

Overall, Topic1 “Plot threads” and Topic5 “Character tips” continue to decline, suggesting that the declining topics may be replaced by new topics. In addition, Topic 3 “Version Updates” reaches the highest level during the saturation period, indicating that users are gradually focusing on version updates and community interactions. Overall, the interests of users in World of Warcraft-related. These findings support H1 by showing that topics evolve over time and differ across platforms. YouTube discussions shift from narrative and promotional content toward update-related topics, while Reddit maintains a stronger focus on technical and gameplay-oriented discussions..

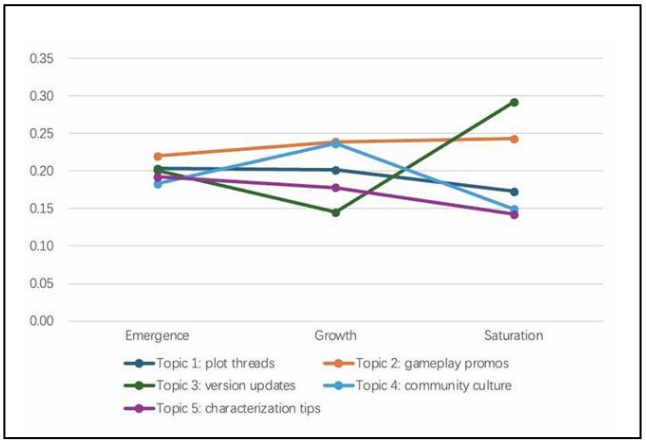


Figure 2. YouTube DTM analysis results

discussions on the YouTube platform have evolved from game promotion and plot lines to version updates and camp culture, showing a diversified discussion pattern over time.

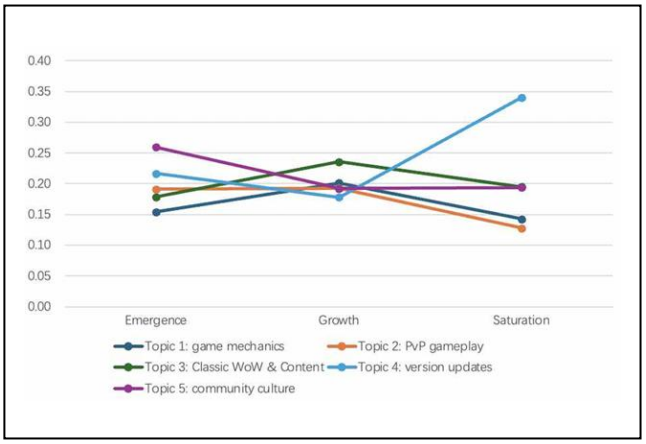


Figure 3. Reddit DTM Analysis Results

Through dynamic topic modeling and line graph analysis of World of Warcraft-related comment data on Reddit platform, the results show the changes of contribution scores of five topics in three time periods.

5.3 Sentiment Analysis

The sentiment distributions differ across platforms. YouTube comments are more concentrated around neutral sentiment values, indicating less polarized emotional expression. In contrast, Reddit exhibits a higher proportion of both strongly positive and strongly negative comments, suggesting more polarized and opinionated discussions. This difference reflects platform characteristics, where YouTube supports broad and immediate emotional expression, while Reddit facilitates more evaluative and discussion-oriented interactions.

5.4 Social Network Analysis

In the study of text and topic communication, SNA is not only applicable to the analysis of user-level interactions but also can be used to model the communication dynamics between “topic nodes”. Specifically, this study focuses on the construction and analysis of Topic Communication Network, constructs topic diffusion paths through semantic similarity and timestamps, and further quantifies the cross-platform influence of each topic using PageRank algorithm. In SNA that fuses YouTube and Reddit comment data, text data from both platforms are first integrated, and topic numbers are assigned to each comment through an LDA topic model. Specifically, the comment data are first binned, and all text data are combined into a large bag-of-words model (Bag-of-Words). Based on this, 10 topics are extracted by training with the LDA model from the gensim library. For each topic, the code extracts the top 10 keywords with the highest weights and their corresponding weights to form a structured Data Frame for subsequent visualization and analysis. Next, the PageRank graph of each node is plotted, as shown in Figure 4.

The PageRank results show that Topic 10 “version update” and Topic 1 “game mechanism” have the highest scores, indicating that they have the most influence in the whole topic dissemination network.

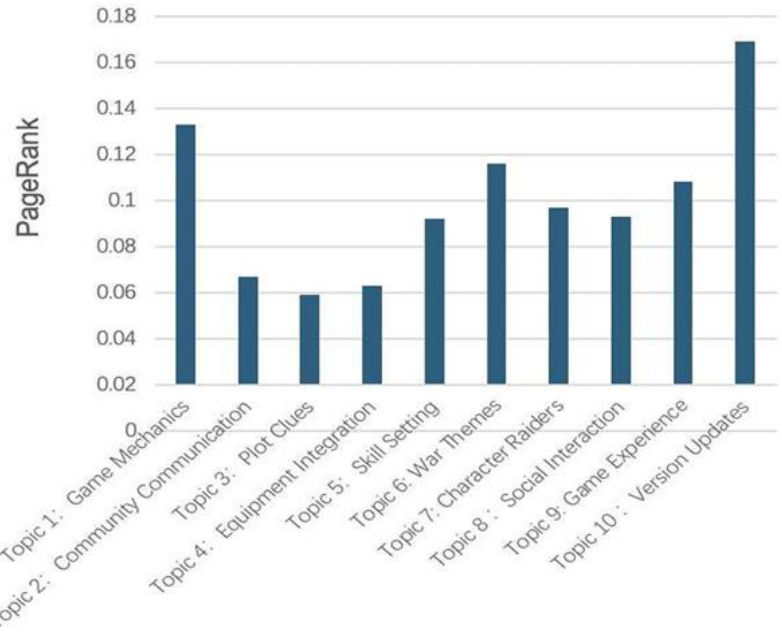


Figure 4. PageRank of Topics

The average propagation delay between YouTube and Reddit is approximately 57 hours. However, variation across topics suggests that diffusion speed is not perfectly uniform. This indicates that different types of content may spread at different rates across platforms. These results support H2 by demonstrating a directional pattern in which topics emerge on YouTube and subsequently appear on Reddit. The PageRank analysis further shows that "version update" and "game mechanism" are the most influential topics in the cross-platform network. It should be noted that topic matching across platforms is based on semantic similarity rather than direct user-level linkage, and therefore reflects diffusion patterns rather than strict causal transmission. It is shown in Figure 5.

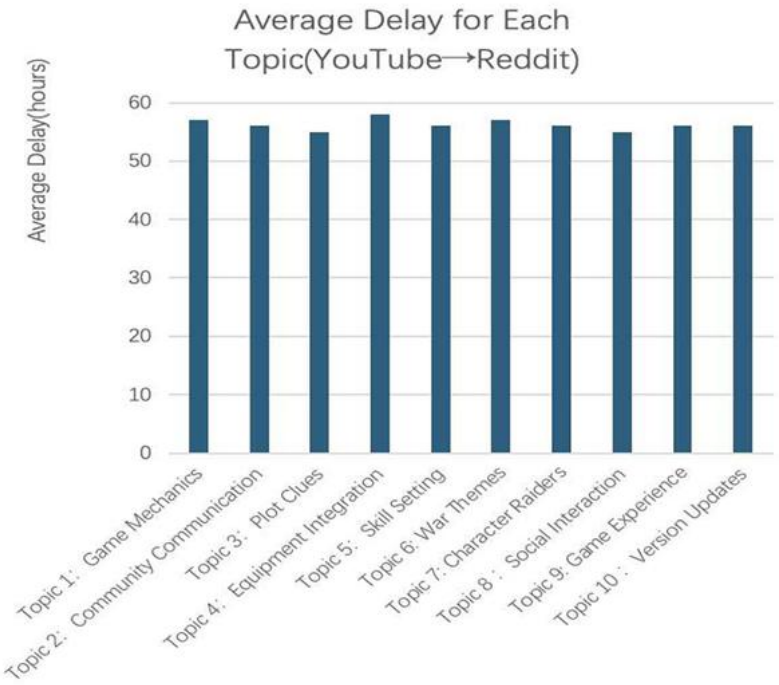


Figure 5. Average Delay for each topic

The MLP model achieves an MSE of 0.25 and an R^2 of 0.67, indicating that YouTube features explain a substantial proportion of variation in Reddit comment volume. The model captures overall trends effectively, particularly during periods of increased discussion activity, although deviations occur during low-activity periods.

As baseline, we compare the MLP model with a univariate autoregressive model using only lagged Reddit comment counts. The MLP model demonstrates improved explanatory power, indicating that incorporating YouTube sentiment and topic intensity provides additional insight beyond internal Reddit dynamics.

These results support H3 by showing that sentiment and topic intensity on YouTube are significantly associated with Reddit discussion activity. Because the model uses contemporaneous features, the findings should be interpreted as explanatory rather than strictly predictive.

Overall, the findings consistently support H1–H3. Topic modeling demonstrates dynamic and platform-specific evolution patterns (H1), network analysis reveals directional cross-platform diffusion (H2), and the ANN model shows a strong association between YouTube signals and Reddit activity (H3).

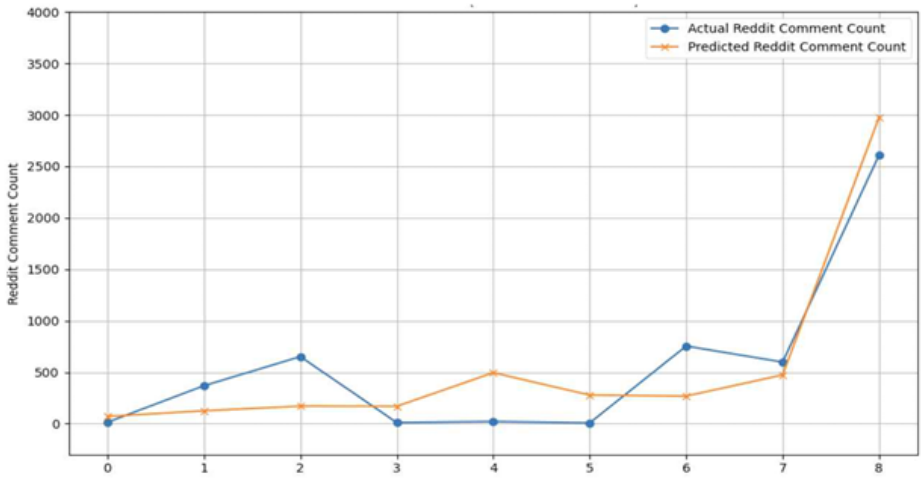


Figure 6. ANN prediction

6 Conclusion

This study examines cross-platform topic and sentiment dynamics between YouTube and Reddit using World of Warcraft as a case. The findings show that discussion patterns differ systematically across platforms and evolve over time. YouTube is characterized by emotionally expressive and high-reach content, while Reddit supports more structured and technical discussions. Topic modeling results demonstrate that user attention shifts from narrative and promotional content toward update-related discussions, while Reddit consistently emphasizes gameplay and problem-solving topics.

The analysis further reveals a directional cross-platform diffusion pattern in which topics tend to emerge on YouTube and subsequently appear on Reddit, with an average delay of approximately 55–60 hours. This suggests that non-specialized platforms play an important role in early-stage attention formation, while specialized communities facilitate deeper interpretation and knowledge exchange.

In addition, the modeling results indicate that sentiment and topic intensity on YouTube are strongly associated with Reddit discussion activity, explaining a substantial proportion of variation in comment volume. These findings highlight the value of cross-platform signals as indicators of community dynamics, although the relationships identified in this study should be interpreted as explanatory rather than strictly predictive.

From an e-business perspective, the results suggest that platform operators, game developers, and digital marketers can leverage early signals from high-reach platforms such as YouTube to monitor emerging topics, anticipate shifts in user attention, and design more timely communication strategies. For example, sentiment trends may help identify potential dissatisfaction following updates, while topic intensity can signal emerging areas of interest for content development and community engagement.

Overall, this study contributes to the understanding of cross-platform communication in digital ecosystems by linking platform characteristics with observable patterns of topic evolution, sentiment expression, and information diffusion. Future research can extend this work by incorporating user-level interaction data, refining diffusion measurement approaches, and applying the framework to other platform ecosystems beyond gaming.

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